

1. Executive Summary

		Traffic Summary		Count	
Metric	Value	Total Flows Observed		19174	
Total Threats Mitigated	64	Total Packets Analyzed		6234258289	
High Priority	13	Malicious Packets		6213998205	
Low Priority	51	True Negatives Passed		11839	
Manual Releases	0				
Manual Blocks	0	Attack Vector		Count	
		ICMP Flood		15	
ML Metrics	Value	SYN Flood		19	
False Positives	92	UDP Flood		10	
FP Rate	0.43%	Uncertain		20	
IF Precision	98.26%				
IF Recall	98.46%	Action		Count	
IF F1-Score	98.36%	Quarantined		10	
IF Accuracy	99.35%	Time Ban		10	
		Blackhole		2	
		Released		10	

2. Performance Benchmark

2a. Isolation Forest: Anomaly Detection

Metric	Value	Description
Precision	98.26%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	98.46%	Share of actual attacks successfully flagged
F1-Score	98.36%	Balanced measure combining Precision and Recall
Accuracy	99.35%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.43%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	1.54%	Actual attacks that went undetected
True Positive Rate (TPR)	98.46%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.57%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 2873	FN 45
Actual: Normal	FP 51	TN 11794

2b. Random Forest: Attack Classification

Metric	Value	Description
Precision	100.00%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	99.97%	Share of actual attacks successfully flagged
F1-Score	99.98%	Balanced measure combining Precision and Recall
Accuracy	99.97%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.00%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	0.03%	Actual attacks that went undetected
True Positive Rate (TPR)	99.97%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	100.00%	Normal traffic correctly identified as safe

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	1996	0	0
Act: ICMP	0	118	0
Act: UDP	0	0	758

2c. Hybrid Cascaded

Metric	Value	Description
Precision	98.26%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	98.42%	Share of actual attacks successfully flagged
F1-Score	98.34%	Balanced measure combining Precision and Recall
Accuracy	99.34%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.43%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	1.58%	Actual attacks that went undetected
True Positive Rate (TPR)	98.42%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.57%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 2872	FN 46
Actual: Normal	FP 51	TN 11794

2d. Response Latency

Metric	Value	Description
Detection Time	302.2 ms	Average time taken to detect an attack
Mitigation Response Time	5.8 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2e. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	3.57%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	11.60%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	6.37%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.17	7	5	2026-09-25 14:48:14	2026-09-25 14:55:35
10.0.0.23	7	5	2026-09-25 14:48:14	2026-09-25 14:55:33
10.0.0.20	6	5	2026-09-25 14:48:16	2026-09-25 14:55:12
10.0.0.16	4	5	2026-09-25 14:48:16	2026-09-25 14:55:12
10.0.0.21	4	5	2026-09-25 14:48:18	2026-09-25 14:55:19
10.0.0.24	4	5	2026-09-25 14:48:15	2026-09-25 14:55:16
10.0.0.25	4	5	2026-09-25 14:48:16	2026-09-25 14:55:20
10.0.0.18	3	5	2026-09-25 14:48:14	2026-09-25 14:52:26

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.19	3	5	2026-09-25 14:48:14	2026-09-25 14:52:26
10.0.0.22	3	2	2026-09-25 14:53:02	2026-09-25 14:55:22
10.0.0.2	4	1	2026-09-25 14:53:00	2026-09-25 14:53:45
10.0.0.5	2	1	2026-09-25 14:53:01	2026-09-25 14:53:24
10.0.0.1	1	1	2026-09-25 14:53:42	2026-09-25 14:53:53
10.0.0.10	1	1	2026-09-25 14:53:18	2026-09-25 14:53:28
10.0.0.7	1	1	2026-09-25 14:53:06	2026-09-25 14:53:16

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-25 14:49:15	10.0.0.23	DDoS	ICMP Flood	91.1%	High	Time Ban (30s)
2026-09-25 14:49:16	10.0.0.23	DDoS	ICMP Flood	91.1%	High	Time Ban
2026-09-25 14:49:17	10.0.0.20	DDoS	UDP Flood	76.3%	High	Time Ban (30s)
2026-09-25 14:49:25	10.0.0.17	DDoS	SYN Flood	92.6%	Critical	Time Ban (1m)
2026-09-25 14:49:25	10.0.0.19	DDoS	SYN Flood	92.2%	Critical	Time Ban (1m)
2026-09-25 14:49:25	10.0.0.18	DDoS	SYN Flood	94.2%	Critical	Time Ban (1m)
2026-09-25 14:49:25	10.0.0.24	DDoS	ICMP Flood	91.0%	High	Time Ban (1m)
2026-09-25 14:49:25	10.0.0.18	DDoS	SYN Flood	94.2%	Critical	Time Ban
2026-09-25 14:49:25	10.0.0.24	DDoS	ICMP Flood	91.0%	High	Time Ban
2026-09-25 14:49:25	10.0.0.17	DDoS	SYN Flood	92.6%	Critical	Time Ban
2026-09-25 14:49:25	10.0.0.19	DDoS	SYN Flood	92.2%	Critical	Time Ban
2026-09-25 14:49:27	10.0.0.16	DDoS	SYN Flood	90.3%	Critical	Time Ban (1m)
2026-09-25 14:49:27	10.0.0.25	DDoS	ICMP Flood	90.7%	Critical	Time Ban (1m)
2026-09-25 14:49:28	10.0.0.21	DDoS	UDP Flood	86.2%	High	Time Ban (1m)
2026-09-25 14:49:45	10.0.0.23	DDoS	ICMP Flood	92.8%	High	Time Ban (1m)
2026-09-25 14:49:47	10.0.0.20	DDoS	UDP Flood	76.3%	High	Time Ban (1m)
2026-09-25 14:50:25	10.0.0.17	DDoS	SYN Flood	92.6%	Critical	Time Ban (2m)
2026-09-25 14:50:25	10.0.0.19	DDoS	SYN Flood	92.2%	Critical	Time Ban (2m)
2026-09-25 14:50:25	10.0.0.18	DDoS	SYN Flood	94.2%	Critical	Time Ban (2m)
2026-09-25 14:50:25	10.0.0.24	DDoS	ICMP Flood	92.6%	High	Time Ban (2m)
2026-09-25 14:50:27	10.0.0.16	DDoS	SYN Flood	90.3%	Critical	Time Ban (2m)
2026-09-25 14:50:27	10.0.0.25	DDoS	ICMP Flood	90.7%	Critical	Time Ban (2m)
2026-09-25 14:50:28	10.0.0.21	DDoS	UDP Flood	86.2%	High	Time Ban (2m)
2026-09-25 14:50:45	10.0.0.23	DDoS	ICMP Flood	93.8%	High	Time Ban (2m)
2026-09-25 14:51:47	10.0.0.20	DDoS	UDP Flood	76.3%	High	Time Ban (2m)
2026-09-25 14:52:26	10.0.0.17	DDoS	SYN Flood	92.6%	Critical	Time Ban (5m)
2026-09-25 14:52:26	10.0.0.19	DDoS	SYN Flood	92.2%	Critical	Time Ban (5m)
2026-09-25 14:52:26	10.0.0.18	DDoS	SYN Flood	94.2%	Critical	Time Ban (5m)
2026-09-25 14:52:26	10.0.0.24	DDoS	ICMP Flood	96.1%	Critical	Time Ban (5m)
2026-09-25 14:52:28	10.0.0.16	DDoS	SYN Flood	90.3%	Critical	Time Ban (5m)
2026-09-25 14:52:28	10.0.0.25	DDoS	ICMP Flood	90.7%	Critical	Time Ban (5m)
2026-09-25 14:52:33	10.0.0.16	DDoS	SYN Flood	90.3%	Critical	Time Ban
2026-09-25 14:52:46	10.0.0.23	DDoS	ICMP Flood	96.6%	Critical	Time Ban (5m)
2026-09-25 14:53:02	10.0.0.22	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 14:53:04	10.0.0.21	DDoS	UDP Flood	86.2%	Critical	Time Ban
2026-09-25 14:53:04	10.0.0.20	DDoS	UDP Flood	76.3%	Critical	Time Ban
2026-09-25 14:53:11	10.0.0.2	DDoS	Uncertain	58.9%	Low	Released

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-25 14:53:12	10.0.0.5	DDoS	Uncertain	58.9%	Low	Released
2026-09-25 14:53:12	10.0.0.22	DDoS	UDP Flood	75.0%	High	Time Ban (30s)
2026-09-25 14:53:12	10.0.0.22	DDoS	UDP Flood	75.1%	Critical	Time Ban
2026-09-25 14:53:16	10.0.0.7	DDoS	Uncertain	48.2%	Low	Released
2026-09-25 14:53:28	10.0.0.10	DDoS	Uncertain	68.7%	Low	Released
2026-09-25 14:53:33	10.0.0.25	DDoS	ICMP Flood	92.2%	Critical	Time Ban
2026-09-25 14:53:53	10.0.0.1	DDoS	Uncertain	61.0%	Low	Released
2026-09-25 14:55:01	10.0.0.16	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 14:55:01	10.0.0.20	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 14:55:02	10.0.0.23	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 14:55:02	10.0.0.17	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 14:55:06	10.0.0.24	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 14:55:08	10.0.0.21	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 14:55:09	10.0.0.25	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 14:55:12	10.0.0.16	DDoS	SYN Flood	92.9%	Critical	Time Ban (10m)
2026-09-25 14:55:12	10.0.0.20	DDoS	UDP Flood	84.3%	Critical	Time Ban (5m)
2026-09-25 14:55:13	10.0.0.23	DDoS	ICMP Flood	89.8%	Critical	Time Ban (10m)
2026-09-25 14:55:13	10.0.0.17	DDoS	Uncertain	-	Low	Released
2026-09-25 14:55:14	10.0.0.23	DDoS	ICMP Flood	89.8%	Critical	Blackhole
2026-09-25 14:55:16	10.0.0.24	DDoS	Uncertain	-	Low	Released
2026-09-25 14:55:19	10.0.0.21	DDoS	Uncertain	-	Low	Released
2026-09-25 14:55:20	10.0.0.25	DDoS	Uncertain	-	Low	Released
2026-09-25 14:55:22	10.0.0.22	DDoS	Uncertain	-	Low	Released
2026-09-25 14:55:25	10.0.0.17	DDoS	SYN Flood	90.0%	Critical	Time Ban (10m)
2026-09-25 14:55:25	10.0.0.18	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 14:55:26	10.0.0.17	DDoS	SYN Flood	90.0%	Critical	Blackhole
2026-09-25 14:55:39	10.0.0.19	DDoS	Uncertain	-	Low	Quarantined

5. IP Attack History (Completed Sessions)

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.17	SYN Flood	0.6323	90.1%	Critical	Phase 3	1	2s	2026-09-25 14:55:35	Escalated to Blackhole
10.0.0.23	ICMP Flood	0.6404	90.4%	Critical	Phase 3	1	0s	2026-09-25 14:55:33	Escalated to Blackhole
10.0.0.17	SYN Flood	0.6350	90.0%	Critical	Phase 3	5	12s	2026-09-25 14:55:26	Escalated to Blackhole
10.0.0.17	SYN Flood	0.6336	90.0%	Critical	Phase 2	5	11s	2026-09-25 14:55:25	Attack persisted - escalating to time ban

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.22	Uncertain	0.0000	-	Low	Phase 1	1	10s	2026-09-25 14:55:22	Attack Stopped
10.0.0.25	Uncertain	0.0000	-	Low	Phase 1	1	11s	2026-09-25 14:55:20	Attack Stopped
10.0.0.21	Uncertain	0.0000	-	Low	Phase 1	1	11s	2026-09-25 14:55:19	Attack Stopped
10.0.0.24	Uncertain	0.0000	-	Low	Phase 1	1	10s	2026-09-25 14:55:16	Attack Stopped
10.0.0.23	ICMP Flood	0.6413	89.8%	Critical	Phase 3	2	12s	2026-09-25 14:55:14	Escalated to Blackhole
10.0.0.17	Uncertain	0.0000	-	Low	Phase 1	1	11s	2026-09-25 14:55:13	Attack Stopped
10.0.0.23	ICMP Flood	0.6388	89.8%	Critical	Phase 2	1	11s	2026-09-25 14:55:13	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6448	84.3%	Critical	Phase 2	1	11s	2026-09-25 14:55:12	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6476	92.9%	Critical	Phase 2	1	11s	2026-09-25 14:55:12	Attack persisted - escalating to time ban
10.0.0.1	Uncertain	0.5989	61.0%	Low	Phase 1	1	11s	2026-09-25 14:53:53	Attack Stopped
10.0.0.20	UDP Flood	0.6478	76.3%	Critical	Phase 2	5	5m 31s	2026-09-25 14:53:47	Escalated to Time Ban
10.0.0.2	Uncertain	0.6092	60.4%	Low	Phase 1	1	10s	2026-09-25 14:53:45	Attack Stopped
10.0.0.22	UDP Flood	0.6318	75.5%	High	Phase 2	2	40s	2026-09-25 14:53:42	Attack persisted - escalating to time ban
10.0.0.2	Uncertain	0.6022	60.4%	Low	Phase 1	1	10s	2026-09-25 14:53:34	Attack Stopped
10.0.0.10	Uncertain	0.6188	68.7%	Low	Phase 1	1	10s	2026-09-25 14:53:28	Attack Stopped
10.0.0.2	Uncertain	0.5920	60.4%	Low	Phase 1	1	11s	2026-09-25 14:53:24	Attack Stopped
10.0.0.5	Uncertain	0.5983	59.1%	Low	Phase 1	1	11s	2026-09-25 14:53:24	Attack Stopped
10.0.0.7	Uncertain	0.6336	48.2%	Low	Phase 1	1	10s	2026-09-25 14:53:16	Attack Stopped
10.0.0.22	UDP Flood	0.6402	75.0%	High	Phase 2	1	10s	2026-09-25 14:53:12	Attack persisted - escalating to time ban
10.0.0.5	Uncertain	0.6010	58.9%	Low	Phase 1	1	11s	2026-09-25 14:53:12	Attack Stopped
10.0.0.2	Uncertain	0.6031	58.9%	Low	Phase 1	1	11s	2026-09-25 14:53:11	Attack Stopped
10.0.0.23	ICMP Flood	0.6200	96.6%	Critical	Phase 2	5	4m 32s	2026-09-25 14:52:46	Escalated to Time Ban
10.0.0.21	UDP Flood	0.6309	86.2%	High	Phase 2	5	4m 11s	2026-09-25 14:52:29	Attack persisted - escalating to time ban
10.0.0.25	ICMP Flood	0.6467	90.7%	Critical	Phase 2	5	4m 12s	2026-09-25 14:52:28	Attack persisted - escalating to time ban

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.16	SYN Flood	0.6497	90.3%	Critical	Phase 2	5	4m 12s	2026-09-25 14:52:28	Attack persisted - escalating to time ban
10.0.0.24	ICMP Flood	0.6191	96.1%	Critical	Phase 2	5	4m 11s	2026-09-25 14:52:26	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6218	94.2%	Critical	Phase 2	5	4m 12s	2026-09-25 14:52:26	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6103	92.2%	Critical	Phase 2	5	4m 12s	2026-09-25 14:52:26	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6154	92.6%	Critical	Phase 2	5	4m 12s	2026-09-25 14:52:26	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6452	76.3%	High	Phase 2	5	3m 31s	2026-09-25 14:51:47	Escalated to Time Ban
10.0.0.20	UDP Flood	0.6452	76.3%	High	Phase 2	5	2m 31s	2026-09-25 14:50:47	Escalated to Time Ban
10.0.0.23	ICMP Flood	0.6244	93.8%	High	Phase 2	5	2m 31s	2026-09-25 14:50:45	Escalated to Time Ban
10.0.0.21	UDP Flood	0.6309	86.2%	High	Phase 2	4	2m 10s	2026-09-25 14:50:28	Attack persisted - escalating to time ban
10.0.0.25	ICMP Flood	0.6467	90.7%	Critical	Phase 2	4	2m 11s	2026-09-25 14:50:27	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6497	90.3%	Critical	Phase 2	4	2m 11s	2026-09-25 14:50:27	Attack persisted - escalating to time ban
10.0.0.24	ICMP Flood	0.6337	92.6%	High	Phase 2	4	2m 10s	2026-09-25 14:50:25	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6283	94.2%	Critical	Phase 2	4	2m 11s	2026-09-25 14:50:25	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6164	92.2%	Critical	Phase 2	4	2m 11s	2026-09-25 14:50:25	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6213	92.6%	Critical	Phase 2	4	2m 11s	2026-09-25 14:50:25	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6452	76.3%	High	Phase 2	4	1m 31s	2026-09-25 14:49:47	Escalated to Time Ban
10.0.0.23	ICMP Flood	0.6336	92.8%	High	Phase 2	4	1m 31s	2026-09-25 14:49:45	Escalated to Time Ban
10.0.0.21	UDP Flood	0.6309	86.2%	High	Phase 2	3	1m 10s	2026-09-25 14:49:28	Attack persisted - escalating to time ban

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.25	ICMP Flood	0.6467	90.7%	Critical	Phase 2	3	1m 11s	2026-09-25 14:49:27	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6497	90.3%	Critical	Phase 2	3	1m 11s	2026-09-25 14:49:27	Attack persisted - escalating to time ban
10.0.0.24	ICMP Flood	0.6365	91.0%	High	Phase 2	3	1m 10s	2026-09-25 14:49:25	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6316	94.2%	Critical	Phase 2	3	1m 11s	2026-09-25 14:49:25	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6215	92.2%	Critical	Phase 2	3	1m 11s	2026-09-25 14:49:25	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6249	92.6%	Critical	Phase 2	3	1m 11s	2026-09-25 14:49:25	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6452	76.3%	High	Phase 2	3	1m 1s	2026-09-25 14:49:17	Escalated to Time Ban
10.0.0.23	ICMP Flood	0.6331	91.1%	High	Phase 2	3	1m 1s	2026-09-25 14:49:15	Escalated to Time Ban

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE